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Article

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Effect of Lithium Ions on Rheology and Interfacial Forces in Ethylammonium Nitrate and Ethanolammonium Nitrate

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Abstract

The effect of added Li^+ to two ionic liquids (ILs), ethylammonium nitrate (EAN) and ethanolammonium nitrate (EtAN) has been investigated using rheology and colloidal probe atomic force microscopy (AFM). Rheology data revealed a complex viscosity dependence that can be ascribed to the different bulk nanostructures. AFM force curves revealed steps for the neat ILs, analogous to those in previous studies. The addition of Li^+ broadened the steps, which is likely an effect of ion clusters formed. Friction measurements corroborate this data and also showed that the structure of EtAN is much more prone to change as Li^+ is added. These results demonstrate the complex behavior of ILs on interfaces and the effect of perturbing such interactions.

Introduction

The nanoscale solid-liquid interface plays an important role in many applications, ranging from lubrication[1-2] to energy conversion and storage.[3] It is the properties of both the solid surface and the liquid which determines their behavior at the interface. Much work has been performed in aqueous solutions to try and understand this interplay,[4-7] especially since solid surface often acquire a charge and thus attracts ions of opposite polarity from the solution. These interactions are described using the Gouy-Chapman-Stern model.[8-10] This model depends on a few assumptions that holds true for simple aqueous systems but fails to account for more complicated liquids, such as ionic liquids (ILs).[11]

ILs consist purely of ions that due to their (large) size and asymmetry have a melting point below 100 ºC.[12] ILs are attracting interest due to their generally large electrochemical windows,[13] high thermal stability,[14] low vapor pressure,[15] and good solvent properties.[12, 14] They can be tuned for different applications by changing the molecular structure of the ions and are thus often referred to as designer solvents.[16] Their charged nature renders them interesting in applications involving the solid-liquid interface, such as lubrication[17-28] and as electrolytes for energy storage.[11, 13, 29] A recent review highlighted the most prevalent problems in IL research and how even the most fundamental models, such as the Gouy-Chapman-Stern, fail to adequately describe the properties of ILs on surfaces.[11] This is because the ion concentration is the same in bulk and at interfaces and their (large and asymmetric) size makes it difficult to model them as point charges. Many studies have shown that ILs have self-assembled nanostructures in bulk[16, 30-32] and on surfaces[16-17, 23-24, 27-28, 33-36] that are in some ways similar to aqueous surfactant structures.[37-40]

The focus of this study is on two protic ILs (formed by proton transfer from a Brønsted acid to a Brønsted base),[41] ethylammonium nitrate (EAN) and ethanolammonium nitrate (EtAN).

Their structures have been studied before in the bulk and on mica surfaces.[21-22, 31, 33, 37-39, 42-50] EAN self-assembles into a sponge-like nanostructure in bulk whereas EtAN forms a clustered morphology, which is less well defined due to reduced solvophobic interactions. The scale of the self-assembly is larger than the ion dimensions and determined by the interplay between electrostatic interactions vs. solvophobic interactions. EAN, for example, has clearly separated polar and apolar domains. The polar domain is formed from the electrostatic interaction between the charged ammonium and nitrate groups. On the other hand, the apolar domain is created from solvophobic interactions between the ethyl chains. Replacing the terminal hydrogen from this chain (EAN) and replacing it with a hydroxide group (EtAN) reduces this interaction significantly and explains why the bulk structuring of EtAN is weaker. The strength of the interaction between the two domains thus determines the self-assembly structure. Interfacial structure is related to that in bulk,[51] but is also influenced by the nature and strength of the ion-surface interactions. In addition to acting as a 2D barrier disrupting the 3D bulk structure there are also topographic implications depending on the local roughness in relationship to the ion sizes.[16]

The effect of adding Li^+ to EAN and EtAN in bulk has been studied before and revealed that even relatively similar ILs react differently in terms of their bulk nanostructure.[52-54] Only small changes were detected for EAN whereas the entire structure changed for EtAN. A recent study demonstrated that the viscosity of EAN increased significantly upon addition of $LiNO_3$ at high fractions (10 and 20 mol%).[55] This was ascribed to the incorporation of Li^+ into the polar domain and its coordination with four anions, thus reducing the mobility and increasing viscosity.

Only a few studies have examined how added Li^+ affects the surface IL properties.[56-59] Added lithium typically results in a reduction in the number of near surface layers, but those that are present are broader.[56-58] When the substrate is anionic, the small hard lithium ion displaces the IL cation and adsorbs to surface charge sites.[58-59]

The focus, and novelty, of this work is to understand how the addition of Li^+ to EAN and EtAN affects different properties of practical interest, such as viscosity, structural forces and friction, in the context of the different self-assembly properties of the two ionic liquids. The results described here show that two apparently rather similar ILs display dramatically different responses upon Li^+ addition.

Materials and methods

The ILs used in this study were synthesized by mixing (a slight excess of) ethylamine (Sigma-Aldrich, USA, 66-72% w/w aqueous solution) or ethanolamine (Sigma-Aldrich, USA, ≥ 99% purity) with nitric acid (RCI Labscan Ltd., Thailand, 70% w/w aqueous solution). To ensure a complete reaction, the nitric acid was diluted further and added drop-wise to the amine solution which was cooled with ice and kept below 5°C (prevents oxide formation). Rotary evaporation (40-50°C for 2h) and heat treatment (100-110°C for 12h) under a N_2-atmosphere were subsequently used to reduce the water content. $LiNO_3$ (Sigma-Aldrich, USA) was dried for at least 2h at 120 °C and then added to the IL solutions and dissolved by ultrasonication (time varied from 10 min to 8h depending on the wt%). The water levels of IL solutions with added $LiNO_3$ were determined by Karl-Fischer titration (Metrohm) before and after experiment and stayed below 0.1 wt% for EAN + $LiNO_3$ and below 0.5 wt% for EtAN + $LiNO_3$. Molecular structures for EAN, EtAN and $LiNO_3$ are found in Figure 1.

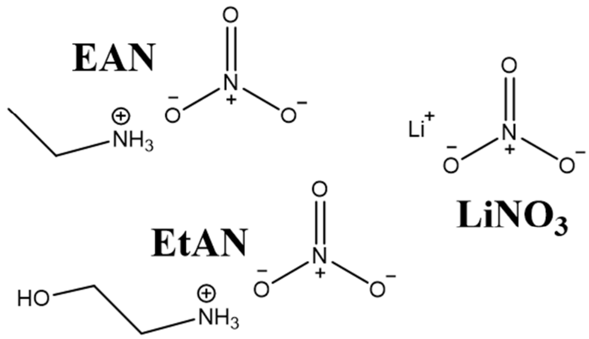

Figure 1. Molecular structures of the investigated ions. Note that nitrate has a resonance structure with a negative charge of 2/3 per oxygen and a net charge of -1.

Rheology measurements were performed on a Malvern Kinexus Rheometer (Worcestershire, UK) between 298 and 353 K. A concentric cylinder configuration was used with a gap of 5.12 mm. The system was allowed to equilibrate for 5 min after each temperature was reached and the shear rate was swept from 1 to 4000 s^{-1}. A value was extracted for each temperature of each solution by averaging the values in the Newtonian range.

Borosilicate particles (Duke Scientific, Palo Alto, CA) with nominal diameters of 14 µm were attached to tipless cantilevers (HQ:NSC36/tipless/No Al, MikroMasch, Tallinn, Estonia) using a two-component epoxy (Selley's Super Strength Araldite). The diameters of attached probes and cantilever dimensions were measured with an optical microscope (Zeiss, Axioskop 40) and determined by using the AxioVision software (Zeiss, Axiovision 4.7). Resonant frequencies and Q-values were obtained by using the cantilever tune function in the AFM software on the neat tipless cantilevers and used together with the cantilever dimensions and probe radii to determine the normal[60] and torsional[61] spring constants.

All colloidal probe[62] AFM experiments were carried out in a sealed fluid cell (silicone O-ring) and all accessories needed (cell, O-ring, connectors and tubes) were treated with a 2 wt% surfactant solution for 30 min and rinsed extensively with reverse osmosis water and absolute ethanol before being dried with nitrogen gas. This was carried out before and after every experiment. Cantilevers were rinsed with reverse osmosis water and absolute ethanol, dried under nitrogen gas and then irradiated with UV/ozone radiation for 10 minutes. The atomically smooth muscovite mica surfaces were cleaved immediately prior to an experiment using adhesive tape. Separate experiments were carried out subsequently for each IL, starting with either neat EAN or EtAN. Each injection volume was 1 ml to ensure complete flushing of the previous concentration from the cell and the system was allowed 30 min to equilibrate before force and friction measurements were carried out.

A Nanoscope IV MultiMode AFM (Bruker, Santa Barbara, CA) equipped with a PicoForce controller and scanner in contact mode (with closed loop enabled) was used for force and friction measurements. Normal force curves were obtained using ramp rates and ramp sizes of 0.02 Hz and 70 nm. The raw data (piezo movement vs. deflection of cantilever) was transformed to force vs. apparent separation according to an established protocol.[63] In general very small ramp sizes were used to minimize the speed. In some cases (e.g. Figure 4C) this meant that no constant compliance region was achieved. In such cases a second force curve was performed using a larger scan size, and the constant compliance value (often referred to as INVOLS) was used to calibrate the spring deflection. The friction force was measured by setting the scan angle to 90° and captured images in contact mode as the load was increased in discrete steps from zero load up to a pre-defined maximum value and then, similarly, decreased until the surfaces detached from each other. The slow scan axis was disabled for these measurements and the friction measurement was performed with a scan rate of 2 Hz and a scan size of 1 μm. Each image consisted of 16 scanned lines in each direction and 256 data points on each line. To extract a friction force at each load the captured images were analyzed. The first and last data points for each line were first removed since they arise from the turning of the probe. Thereafter, the captured images from the backwards motion were subtracted from the forward motion and the resulting data points averaged to extract values of the difference in friction signal, ΔV_{fric}. These values were then transformed to friction forces at each load, F_{fric}, according to the following equation:

$$F_{fric} = \frac{\Delta V_{fric} k_{\phi}}{2 h_{eff} \delta} \quad (1)$$

where k_{ϕ} is the torsional spring constant, h_{eff} the diameter of the probe plus half the thickness of the cantilever and δ the torsional detector sensitivity.[64] The friction data was processed using the FrictionIT v.2.6.4 software (ForceIT, Sweden).

Results and discussion

Rheology

In Figure 2, shear viscosities are presented in Arrhenius plots for different Li^+ concentrations in EAN and EtAN. There is a linear dependence on inverse temperature for all investigated solutions, indicating Arrhenius type behavior (single activation energy). The shear viscosities of neat EAN and EtAN are in good agreement with previous papers.[46, 65-66] For EAN only very small changes (<1 %) in shear viscosity are measured for low concentrations of $LiNO_3$ (≤1 wt%). At higher concentrations a more significant change is observed. The measured viscosity is around 70 mPa s for 10 wt% $LiNO_3$ in EAN compared to 40 mPa s for neat EAN. The measured viscosity values for EtAN reveal a different dependence on $LiNO_3$ concentration. Even at 0.1 wt% there is a difference (~10 %) in viscosity compared to neat EtAN and at 10 wt% the viscosity is 2.5 times larger than for neat EtAN.

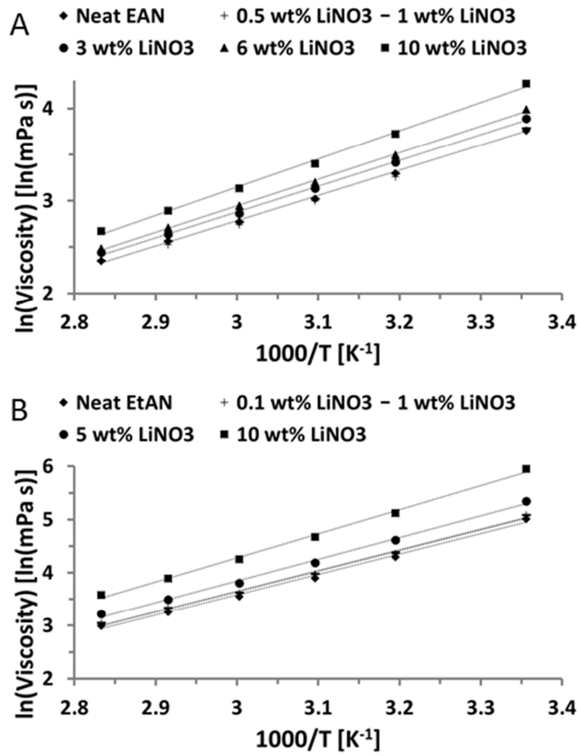

Figure 2. Measured shear viscosity plotted as a function of temperature and lithium concentration in Arrhenius form for A) EAN and B) EtAN. The lines are linear fits to the data with R^2>0.995. Note that the data for neat EAN, 0.5 and 1 wt% $LiNO_3$ as well as 0.1 and 1 wt% $LiNO_3$ in EtAN are indistinguishable from each other.

The strong effect of Li^+ addition to EtAN and weak effect for EAN is supported by structural studies.[52, 54] Neutron diffraction and empirical potential structure refinement modelling as well as molecular dynamics were used to elucidate the bulk nanostructure of lithium ions dissolved in EAN and EtAN (only Ref[54]). At low concentrations of Li^+ (1:30) in EAN, only minuscule changes were found in the nanostructure. This was interpreted as Li^+ only affecting the packing but not ordering. Furthermore, it was shown that Li^+ primarily dissolves in the polar domain and interacts strongly with the anion. Small amounts of Li^+ in EAN will thus only marginally affect bulk properties such as viscosity. Larger concentrations, on the other hand, do reveal an increase in viscosity so there must be a transition at which the ordering or packing of EAN is (at least somewhat) distorted or that larger aggregates are formed. The bulk nanostructure studies did also investigate higher Li^+ concentrations in EAN and found that the apolar domain was slightly perturbed due to a decrease in strength for the solvophobic interactions between ethyl groups, which was caused by the presence of $LiNO_3$.[52, 54] Moreover, Li^+ coordinated to approximately 4 nitrate ions with short bond lengths (<2 Å) in the polar domain. The short bond lengths indicate a solid-like structure within the domain and the large viscosity increase measured here for 10 wt% $LiNO_3$ in EAN is presumably due to these structures.

EtAN, however, changes its bulk nanostructure from a clustered morphology to sponge-like on addition of Li^+ but the coordination number, location of Li^+ in the polar domain and bond length between Li^+ and NO_3^- are all similar to EAN when the concentration of Li^+ was high.[54] The nanostructure of EtAN has, like EAN, polar and nonpolar domains but they are not networked as strongly as for EAN due to the hydroxide group that interacts equally well with both domains. The polar domain is therefore more accessible for Li^+ in EtAN and forces the hydroxide group to interact primarily with other hydroxide groups and the alkyl chains therefore associates. EtAN therefore changes its structure more easily, which affects bulk properties such as viscosity. The more significant viscosity increase for EtAN can therefore be explained by the different bulk nanostructure. However, how this translates to the surface properties of the systems is unclear.

Structural forces

EAN and EtAN have well defined nanostructures at interfaces that are much flatter than the bulk structures.[33, 37, 45, 47] Upon confinement this interfacial structure is “squeezed out” which produced step-like contact mode force curves. The maximum force associated with each step increases with decreasing separation, which produces an oscillatory force profile with the periodicity between each oscillation corresponding to the ion-pair diameter. The high viscosity of ILs, however, renders them difficult to measure compared to many other molecular liquids so the relative velocities have to be greatly reduced. Here, structural forces were measured for the neat ILs and ILs with 1 or 10 wt% added $LiNO_3$ to investigate how the addition of Li^+ affects the boundary layer. In Figure 3A, data for measured force against apparent separation is presented and two steps are observed for neat EAN. The width between the steps is in agreement with the ion-pair distance of EAN.[33] Furthermore, the force required to penetrate the steps increase with decreasing apparent separation, suggesting enhanced (flatter) structure closer to the surface. Also, the slope of the outermost step is much lower than the innermost steps, indicating increased compressibility. The innermost step at 0.3 nm appears to be shorter than the ion-pair distance of EAN. It has previously been shown that the cation is very firmly adsorbed to the mica, and remains in place even at large applied forces with sharp tips.[37-38] The squeeze out at the innermost step is therefore likely cations adsorbed to the probe (slight negative charge on silica) being squeezed out of contact.[22, 66]

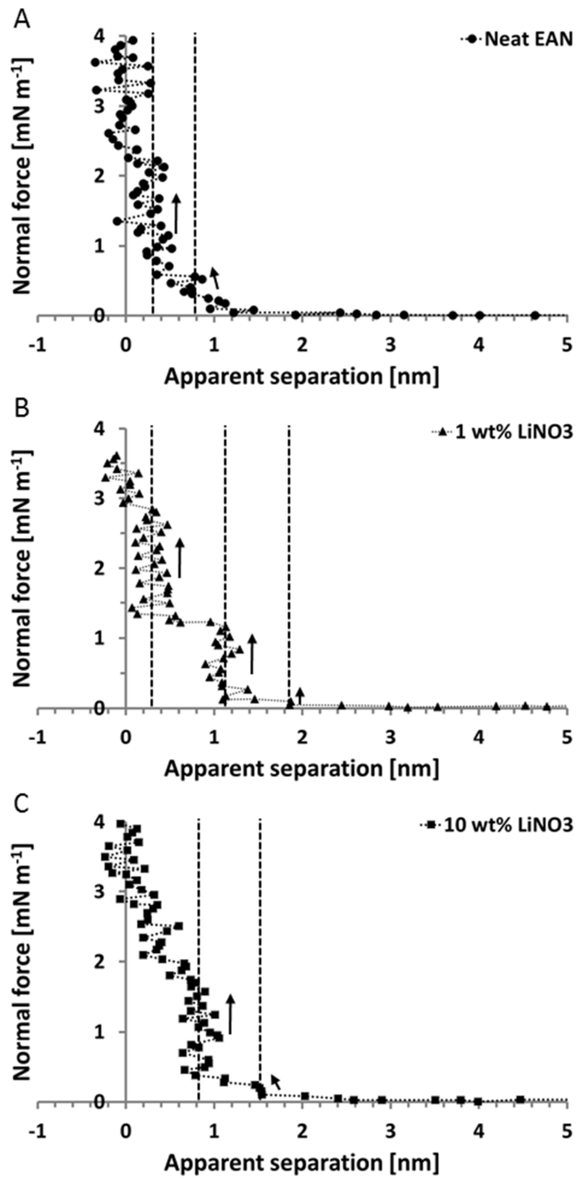

Figure 3. Force profiles showing structural barriers at approach for A) neat EAN, B) 1 wt% $LiNO_3$, and C) 10 wt% $LiNO_3$. Dotted lines are a guide to the eye and the arrow indicate the compressibility at each step.

The addition of 1 wt% $LiNO_3$ to EAN introduces small changes to the force profile (Figure 3B). The width of the outermost step has broadened from 0.5 nm to 0.8 nm. It is likely an ordering effect of Li^+ incorporated into the near surface structure since Li^+ is small and has a localized charge. Li^+ has an ionic radius of 0.59 Å (for coordination number 4)[67] and is known to interact strongly with nitrate ions.[52, 54] The strength of the barrier is also increased and points toward a more ordered structure being squeezed out of contact. At the innermost step, the width remains unchanged but a larger repulsive force is required to penetrate the step. This can be caused by the lithium ions inducing better packing of the cations adsorbed on the surface.

At 10 wt% $LiNO_3$ in EAN the widths of the outermost steps are consistent with those found at 1 wt% (Figure 3C). The force required to penetrate the interfacial structure has increased and the outermost "step" at an apparent separation of 1.55 nm is now clearer. An increased concentration of Li^+ has thus introduced more order to the interfacial structure. The rheology measurements demonstrated that the shear viscosity at this concentration was increased significantly (Figure 2). This could be why there is a gradual increase in force from the innermost step into apparent contact as it could be interpreted as viscous hindrance. Also, the absence of a small step at short apparent separations suggests that the adsorbed layer have changed.

Analogous force profiles "sensing" interfacial structure for EtAN solutions are shown in Figure 4. Two repulsive barriers appear for neat EtAN (Figure 4A), each 0.55-0.6 nm in width, which is in good agreement with the ion-pair distance (0.55 nm).[47] Unlike for EAN, the innermost step in the force profile is consistent with a full ion-pair being squeezed out of contact. This is presumably because EtAN is less amphiphilic.[38] However, the maximum applied force is lower for EtAN compared to EAN so it could also be that the apparent contact is wider, and that additional steps would be observed at higher force. An increase in viscosity would induce viscous hindrance as the two surfaces approach each other but, notably, there is no such gradual increase observed here. EtAN is more viscous than 10 wt% $LiNO_3$ in EAN so the absence of thereof suggests that something else is responsible for its occurrence.

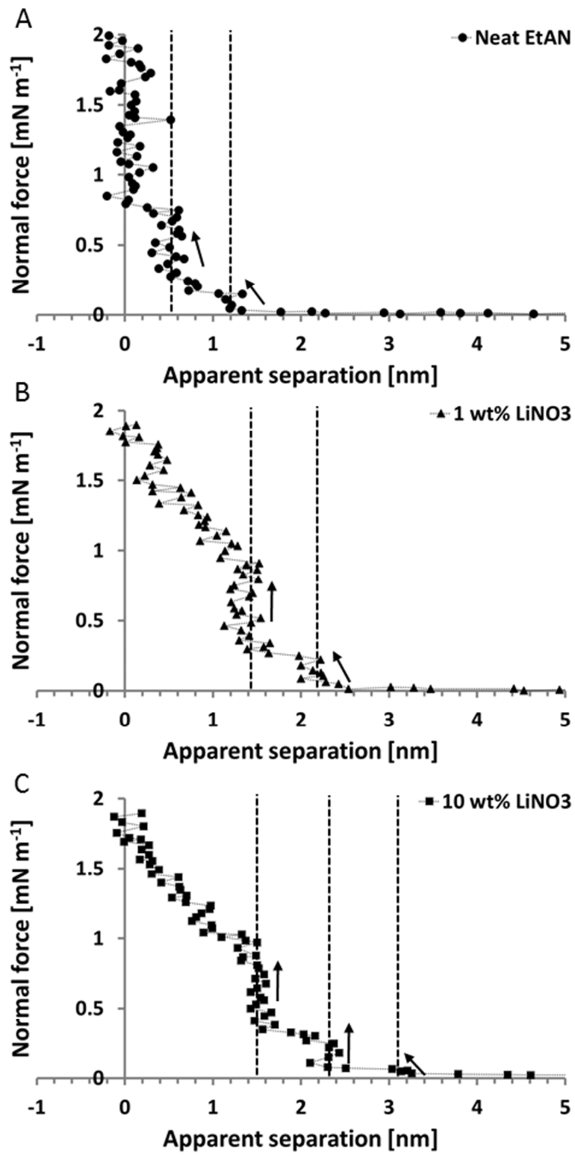

Figure 4. Force-distance curves showing structural barriers for A) neat EtAN, B) 1 wt% $LiNO_3$, and C) 10 wt% $LiNO_3$. Dotted lines are a guide to the eye and arrows indicate the compressibility of the barrier.

At 1 wt% $LiNO_3$ in EtAN two steps are measured, with a distance of 0.8 nm separating them (Figure 4B). The step width is thus similar to that observed for Li^+ in EAN, and supports lithium ions forming larger ionic structures close to the surface. Furthermore, at an apparent separation of 1.4 nm there is a gradual increase in force, similar to what was measured for 10 wt% $LiNO_3$ in EAN. This suggests that the salt is (at least partly) responsible for this compression. There could also be a small step at an apparent separation of 0.4 nm but it is difficult to say from this data. As the concentration of Li^+ was increased further, a similar force profile emerges but there are differences (Figure 4C). The compression is larger and there is no step measured at short separation, which could suggest a complete removal of an adsorbed Li^+ layer. A third step is measured at an apparent separation of 3.2 nm and indicates that the ordering has increased. The width of the steps are again 0.8 nm and further supports larger ionic networks being formed in the near surface structure.

Friction forces

Comparison of Figure 3 and Figure 4 reveals that the addition of Li^+ affects the interfacial structure more for EtAN than EAN. This is attributed to the weaker structure of EtAN, which makes it easier for Li^+ to incorporate itself into the polar domain as well as disrupting interfacial structure. Friction measurements were performed to demonstrate these changes in the boundary layer. Figure 5 shows data for frictional forces measured as a function of normal forces for all investigated systems. Similarly sized probes were used for EAN and EtAN but the spring constant was different so applied normal forces are higher for EAN. The hysteresis on loading/unloading was small in both A and B so the data was averaged for clarity. Neat EAN shows behavior analogous to previous studies.[21-22, 66] The graph reveals two distinct regions separated by a transition. Low loads corresponds to the near surface structure in Figure 3A and the measured friction there is low. As the probe penetrates the innermost step in Figure 3A there is a transition in friction force due to the breaking of solvophobic interactions between cation alkyl chains. At loads larger than 2 mN m^{-1} the measured friction force increases linearly (friction coefficient is ~0.5) with applied load and the probe presumably slides on the remaining layer of adsorbed cation.[21-22] The measured friction forces for 1 wt% $LiNO_3$ in EAN are comparable with those of neat EAN. At large loads the friction forces overlap well and indicates a similar surface composition. It is therefore unlikely that

Li^+ has replaced EAN cations at the surface for this concentration. The friction forces are, however, larger at lower loads and suggests a different near surface structure. This correlates with the changes observed in the force profiles in Figure 3, where the steps were broader and had an increased strength. It was proposed that the addition of Li^+ allows the formation of larger ionic structures and it is presumably those that are now causing an increase in friction force. The linearity at high loads in Figure 5 also starts at a slightly larger load for 1 wt% $LiNO_3$ in EAN compared to the neat case, which is consistent with an increase in order close to the surface. This further supports that as soon as these structures are removed from contact the sliding layer is of similar composition.

At 10 wt% $LiNO_3$ in EAN the measured friction forces reveal a different behavior. At low loads the absolute friction force is larger than it was at 1 wt% $LiNO_3$ in EAN and thus supports the formation of more ordered ionic structures. Furthermore, the friction force is consistently larger at higher loads and with an increased friction coefficient compared to neat EAN and 1 wt% $LiNO_3$ in EAN (from ~0.5 to ~0.7). This demonstrates that the sliding layer is different and that EAN cations are likely absent, corroborating the force profile in Figure 3C. Structural and frictional forces thus reveals that small concentrations of Li^+ only changes the near surface composition of EAN. However, as the Li^+ content is increased significantly there are enough lithium ions to disrupt the adsorbed layer as well.

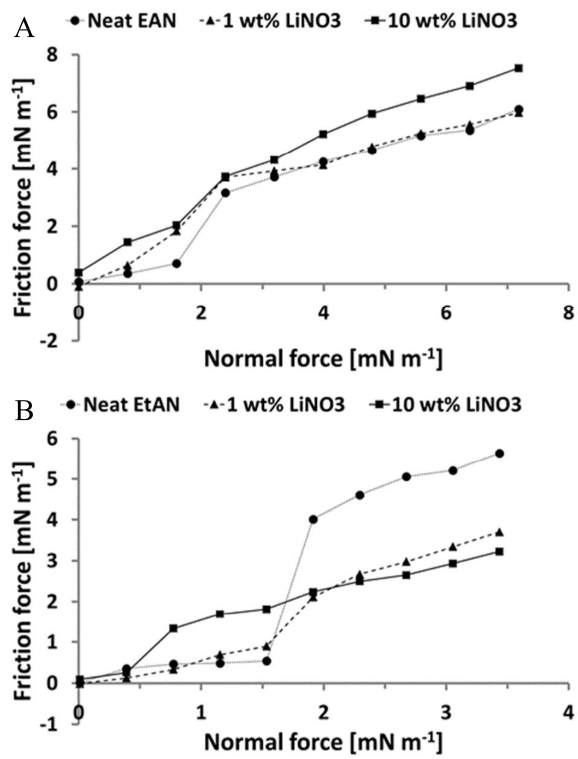

Figure 5. Friction forces measured for A) EAN solutions and B) EtAN solutions as a function of applied load.

The data from friction measurements of neat EtAN and the addition of 1 and 10 wt% $LiNO_3$ to EtAN are shown in Figure 5B. Neat EtAN also shows a two region behavior, analogous to neat EAN. However, the point of transition corresponds to apparent contact in Figure 4A and suggests that zero separation is actually reached at a larger applied load. The introduction of 1 wt% $LiNO_3$ in EtAN reveal similar friction forces to neat EtAN at lower loads but a much reduced absolute force at large loads. The transition between the two regions corresponds to the suggested step at an apparent separation of 0.4 nm in Figure 4B. However, the transition is much less sharp than for neat EtAN and suggests that the energy penalty is less for squeezing out liquid. This demonstrates that the surface composition has changed either at the surface, close to the surface or both. The friction forces at low loads are too low to reveal whether other transitions are present for the steps further away from the surface.

At the high-load regime the friction force is consistently smaller for 1 wt% $LiNO_3$ in EtAN and the boundary layer must therefore have changed. Either there is direct contact between the probe and surface or lithium ions that have adsorbed to the surface. At 10 wt% $LiNO_3$ in EtAN the friction is similar to 1 wt% $LiNO_3$ at high loads and consistent with a much lower absolute friction force compared to neat EtAN. For the low load regime, though, the onset occurs at a lower load and the friction is higher compared to what it was for 1 wt% $LiNO_3$ in EtAN. At that concentration, the transition was attributed to a small step at short apparent separations but this layer is absent for the higher concentration (Figure 4B-C). The transition instead correlates with the step at an apparent separation of 1.55 nm before the gradual increase in force. This step has a similar strength for both 1 and 10 wt% $LiNO_3$ in EtAN but the measured friction force is clearly different. There is currently no explanation for this. The boundary layer of EtAN appear to be more sensitive to the addition of Li^+, presumably due to its weaker nanostructure but the question is how sensitive it actually is. To test the strength of the lithium ion effect in EtAN, a solution was diluted so that it consisted of 0.001 wt% $LiNO_3$ in EtAN and a similar friction experiment was performed on that system. The result is shown in Figure S1 (ESI) and reveal that even very small concentrations of lithium result in a reduced friction force. This indicates that the process is ubiquitous in EtAN.

Conclusions

The adsorption and rheological behavior of Li^+ in two very similar ILs is clearly different. EAN is weakly dependent, with only a small change in viscosity, structural forces and friction at low Li^+ concentrations. Nonetheless, the addition of Li^+ introduces the formation of larger ionic structures in the near surface structure. This results in a slight increase in friction in this region but at larger loads the friction forces overlap with neat EAN. There is a threshold in the lithium ion concentrations where the results deviate. At larger concentrations the viscosity increases and a change in composition occurs only in the near surface layers but also in the adsorbed layers. This has consequences on the friction performance, which demonstrates an increase in friction force at all loads.

EtAN's behavior is strongly dependent on lithium ion concentration. At low concentrations, the viscosity has increased and the structural forces and associated friction profiles are deviating from neat EtAN. The surface composition has changed not only in the near surface layers but also on the surface. This has consequences on the boundary layer as was demonstrated in the large reduction in measured friction forces at high loads. High Li^+ concentrations reveal a similar but larger change. Due to the strong effect of Li^+ on EtAN a tiny concentration of $LiNO_3$ (0.001 wt%) was investigated and demonstrated that even at such small concentrations there was a large reduction in friction force. EtAN is clearly much more prone to changes in its nanostructure than EAN, which can be explained from its bulk nanostructure. In EtAN the polar and nonpolar domains are less associated and any added ion will therefore easier access the polar domains. These results demonstrate that small changes in ion structures are enough to change and as a consequence control the surface properties of a system.

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Supporting information available:

Figure showing measured friction forces for 0.001 wt% $LiNO_3$ in EtAN. This

material is available free of charge via the Internet at http:// pubs.acs.org.

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